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Hydrodynamically actuated preservative free dispensing system

Abstract

Multi-dose preservative-free ocular fluid delivery devices are provided. The fluid delivery device includes a fluid dispensing system and a fluid package for storing a liquid therein and supplying said liquid to the dispensing system. The dispensing system comprises an elongated chamber which includes a check valve which defines a frontal closure to the chamber. The valve is normally closed and hermetically seals the chamber. The dispenser includes a vibration motor that induces oscillations to the chamber and to the fluid within. The oscillations of the chamber impart momentum to the fluid stored in the chamber which in turn imparts force that cyclically opens the valve to dispense streams or liquid droplets. Fluid is dispensed only when the motor oscillates while otherwise the valve is hermetically closed.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
3640274	12/1971	Costello	128/200.14	A61H 35/02
3779245	12/1972	Windsor	N/A	N/A
3812854	12/1973	Michaels et al.	N/A	N/A
3861386	12/1974	Harris et al.	N/A	N/A
3934585	12/1975	Maurice	N/A	N/A
3970250	12/1975	Drews	N/A	N/A
3976072	12/1975	Walker	N/A	N/A
4159803	12/1978	Cameto et al.	N/A	N/A
4300546	12/1980	Kruber	N/A	N/A
4334531	12/1981	Reichl et al.	N/A	N/A
4338576	12/1981	Takahashi et al.	N/A	N/A
4344744	12/1981	Schuster et al.	N/A	N/A
4352459	12/1981	Berger et al.	N/A	N/A
4465234	12/1983	Maehara et al.	N/A	N/A
4632311	12/1985	Nakane et al.	N/A	N/A
4655393	12/1986	Berger	N/A	N/A
4793339	12/1987	Matsumoto et al.	N/A	N/A
4850534	12/1988	Takahashi et al.	N/A	N/A
4882150	12/1988	Kaufman	N/A	N/A
4952581	12/1989	Bito et al.	N/A	N/A
4961345	12/1989	Tsuruoka et al.	N/A	N/A
4976259	12/1989	Higson et al.	N/A	N/A
4981625	12/1990	Rhim et al.	N/A	N/A
5025957	12/1990	Ranalletta et al.	N/A	N/A
5171306	12/1991	Vo	N/A	N/A
5232363	12/1992	Meller	N/A	N/A
5368582	12/1993	Bertera	N/A	N/A
5370317	12/1993	Weston	N/A	N/A

5487378	12/1995	Robertson et al.	N/A	N/A
5549249	12/1995	Foster et al.	N/A	N/A
5624057	12/1996	Lifshey	N/A	N/A
5627611	12/1996	Scheiner	N/A	N/A
5630793	12/1996	Rowe	N/A	N/A
5657926	12/1996	Toda	N/A	N/A
5692651	12/1996	Fuchs	N/A	N/A
5811443	12/1997	DeSantis, Jr. et al.	N/A	N/A
5828394	12/1997	Khuri-Yakub et al.	N/A	N/A
5938117	12/1998	Ivri	N/A	N/A
5958342	12/1998	Gamble et al.	N/A	N/A
5960224	12/1998	Sanada et al.	N/A	N/A
6024717	12/1999	Ball et al.	N/A	N/A
6062212	12/1999	Davison et al.	N/A	N/A
6065623	12/1999	Hierzer et al.	N/A	N/A
6095376	12/1999	Hennemann et al.	N/A	N/A
6196218	12/2000	Voges	N/A	N/A
6232129	12/2000	Wiktor	N/A	N/A
6273092	12/2000	Nolan	N/A	N/A
6302101	12/2000	Py	128/200.22	B05B 11/1066
6467476	12/2001	Ivri et al.	N/A	N/A
RE38077	12/2002	Cohen et al.	N/A	N/A
6543442	12/2002	Gonda et al.	N/A	N/A
6629646	12/2002	Ivri	N/A	N/A
6730066	12/2003	Bennwik et al.	N/A	N/A
6758837	12/2003	Péclat et al.	N/A	N/A
6869275	12/2004	Dante et al.	N/A	N/A
7066398	12/2005	Borland et al.	N/A	N/A
7105357	12/2005	Kalkum et al.	N/A	N/A
7201732	12/2006	Anderson et al.	N/A	N/A
7314938	12/2007	Shen et al.	N/A	N/A
7571722	12/2008	Wuttke et al.	N/A	N/A
7745460	12/2009	Shen et al.	N/A	N/A
7790743	12/2009	Shen et al.	N/A	N/A
7874467	12/2010	Pardes et al.	N/A	N/A
7883031	12/2010	Collins, Jr. et al.	N/A	N/A
7928122	12/2010	Shen et al.	N/A	N/A
8012136	12/2010	Collins, Jr. et al.	N/A	N/A
8048047	12/2010	Domash	N/A	N/A
8056766	12/2010	Grevin	N/A	N/A
8128606	12/2011	Anderson et al.	N/A	N/A
8133210	12/2011	Al-Abdulla et al.	N/A	N/A
8144399	12/2011	Steenblik et al.	N/A	N/A
8168655	12/2011	Gadek et al.	N/A	N/A
8273307	12/2011	Eickhoff et al.	N/A	N/A
8367701	12/2012	Burnier et al.	N/A	N/A
8398001	12/2012	Borland et al.	N/A	N/A
8435544	12/2012	Mitra et al.	N/A	N/A
8544462	12/2012	Papania et al.	N/A	N/A

8545463	12/2012	Collins, Jr. et al.	N/A	N/A
8592450	12/2012	Gadek et al.	N/A	N/A
8629111	12/2013	Acheampong et al.	N/A	N/A
8633162	12/2013	Acheampong et al.	N/A	N/A
8642556	12/2013	Acheampong et al.	N/A	N/A
8648048	12/2013	Acheampong et al.	N/A	N/A
8684980	12/2013	Hunter et al.	N/A	N/A
8685930	12/2013	Acheampong et al.	N/A	N/A
8722728	12/2013	Wong et al.	N/A	N/A
8733935	12/2013	Ballou, Jr. et al.	N/A	N/A
8863998	12/2013	Painchaud et al.	N/A	N/A
8927574	12/2014	Burnier	N/A	N/A
8927921	12/2014	Nelms et al.	N/A	N/A
8936021	12/2014	Collins, Jr.	N/A	N/A
9039666	12/2014	Voss et al.	N/A	N/A
9068566	12/2014	Ivri	N/A	N/A
9085553	12/2014	Zeller et al.	N/A	N/A
9087145	12/2014	Ballou, Jr. et al.	N/A	N/A
9186690	12/2014	Scanlon et al.	N/A	N/A
9216174	12/2014	Shen et al.	N/A	N/A
9238532	12/2015	Decock et al.	N/A	N/A
9248191	12/2015	Acheampong et al.	N/A	N/A
9353088	12/2015	Burnier	N/A	N/A
9447077	12/2015	Burnier et al.	N/A	N/A
9597230	12/2016	Haffner et al.	N/A	N/A
9676525	12/2016	Greiner-Perth et al.	N/A	N/A
9700686	12/2016	Gavini et al.	N/A	N/A
9801757	12/2016	Voss et al.	N/A	N/A
9808825	12/2016	Aguilar et al.	N/A	N/A
9867933	12/2017	Pardes et al.	N/A	N/A
9890141	12/2017	Burnier	N/A	N/A
10073949	12/2017	Ballou, Jr. et al.	N/A	N/A
10105720	12/2017	Decock et al.	N/A	N/A
10124000	12/2017	Shen et al.	N/A	N/A
10154923	12/2017	Hunter et al.	N/A	N/A
10174017	12/2018	deLong et al.	N/A	N/A
10314740	12/2018	Kraft	N/A	N/A
10624781	12/2019	Ivri	N/A	N/A
11278448	12/2021	Palanker et al.	N/A	N/A
11925577	12/2023	Ivri	N/A	B05B 1/083
11938057	12/2023	Ivri	N/A	N/A
2001/0035184	12/2000	Schuler et al.	N/A	N/A
2001/0036424	12/2000	Takahashi et al.	N/A	N/A

2001/0036449	12/2000	Garst	N/A	N/A
2002/0078947	12/2001	Gumaste	N/A	N/A
2002/0124843	12/2001	Skiba et al.	N/A	N/A
2002/0158196	12/2001	Berggren et al.	N/A	N/A
2002/0161344	12/2001	Peclat et al.	N/A	N/A
2002/0185125	12/2001	Klimowicz et al.	N/A	N/A
2002/0190079	12/2001	Hamamoto	N/A	N/A
2003/0052573	12/2002	Wischnewskiy	N/A	N/A
2003/0065294	12/2002	Pickup et al.	N/A	N/A
2003/0071071	12/2002	Garcia et al.	N/A	N/A
2003/0140921	12/2002	Smith et al.	N/A	N/A
2004/0039355	12/2003	Gonzalez et al.	N/A	N/A
2004/0050861	12/2003	Lisec et al.	N/A	N/A
2004/0138630	12/2003	Al-Abdulla et al.	N/A	N/A
2004/0163645	12/2003	Connelly et al.	N/A	N/A
2004/0173642	12/2003	Clifford et al.	N/A	N/A
2004/0204674	12/2003	Anderson et al.	N/A	N/A
2004/0215157	12/2003	Peclat et al.	N/A	N/A
2004/0256487	12/2003	Collins, Jr. et al.	N/A	N/A
2004/0263567	12/2003	Hess et al.	N/A	N/A
2005/0001981	12/2004	Anderson et al.	N/A	N/A
2005/0006417	12/2004	Nicol et al.	N/A	N/A
2005/0107832	12/2004	Bernabei	N/A	N/A
2005/0172962	12/2004	Gumaste et al.	N/A	N/A
2005/0207917	12/2004	Koerner et al.	N/A	N/A
2005/0214129	12/2004	Greene et al.	N/A	N/A
2005/0240162	12/2004	Chen et al.	N/A	N/A
2005/0261641	12/2004	Warchol et al.	N/A	N/A
2006/0065677	12/2005	Py et al.	N/A	N/A
2006/0069358	12/2005	Gerondale	N/A	N/A
2006/0147313	12/2005	Zengerle et al.	N/A	N/A
2006/0210604	12/2005	Dadey et al.	N/A	N/A
2007/0051362	12/2006	Sullivan et al.	N/A	N/A
2007/0088267	12/2006	Shekalim	N/A	N/A
2007/0088268	12/2006	Edwards	N/A	N/A
2007/0102455	12/2006	Stark et al.	N/A	N/A
2007/0119969	12/2006	Collins, Jr. et al.	N/A	N/A
2007/0195151	12/2006	Anderson et al.	N/A	N/A
2007/0268340	12/2006	Dacquay et al.	N/A	N/A
2007/0295332	12/2006	Ziegler et al.	N/A	N/A
2008/0039807	12/2007	Pine	N/A	N/A
2008/0202514	12/2007	Kriksunov et al.	N/A	N/A
2008/0214940	12/2007	Benaron et al.	N/A	N/A
2008/0233053	12/2007	Gross et al.	N/A	N/A
2008/0247264	12/2007	Gabl et al.	N/A	N/A
2008/0257911	12/2007	Choi et al.	N/A	N/A
2009/0060793	12/2008	Eickhoff et al.	N/A	N/A
2009/0114742	12/2008	Collins, Jr.	N/A	N/A
2009/0134235	12/2008	Ivri	N/A	N/A
2009/0182291	12/2008	Eilat	N/A	N/A

2009/0192443	12/2008	Collins, Jr.	N/A	N/A
2009/0212127	12/2008	Reynolds et al.	N/A	N/A
2009/0212133	12/2008	Collins, Jr.	239/338	A61M 11/005
2009/0223513	12/2008	Papania et al.	N/A	N/A
2010/0001090	12/2009	Neergaard et al.	N/A	N/A
2010/0005903	12/2009	Beavis	N/A	N/A
2010/0013352	12/2009	Pletner et al.	N/A	N/A
2010/0044460	12/2009	Sauzade	N/A	N/A
2010/0072301	12/2009	Cater	N/A	N/A
2010/0072302	12/2009	Cater	N/A	N/A
2010/0076388	12/2009	Cater	N/A	N/A
2010/0147899	12/2009	Nardi	N/A	N/A
2010/0186738	12/2009	Kobayashi et al.	N/A	N/A
2010/0222752	12/2009	Collins, Jr. et al.	N/A	N/A
2010/0236545	12/2009	Kern	N/A	N/A
2010/0295420	12/2009	Wierach	N/A	N/A
2010/0326431	12/2009	Yu	N/A	N/A
2011/0074247	12/2010	Hohlfeld et al.	N/A	N/A
2011/0102735	12/2010	Gupta et al.	N/A	N/A
2011/0106025	12/2010	Hall et al.	N/A	N/A
2011/0146670	12/2010	Gallem et al.	N/A	N/A
2011/0203580	12/2010	Papania et al.	N/A	N/A
2011/0284579	12/2010	Pardes et al.	N/A	N/A
2011/0293452	12/2010	Kim et al.	N/A	N/A
2011/0305425	12/2010	Fabrykowski et al.	N/A	N/A
2012/0017898	12/2011	Moller	N/A	N/A
2012/0062840	12/2011	Ballou, Jr. et al.	N/A	N/A
2012/0070467	12/2011	Ballou, Jr. et al.	N/A	N/A
2012/0143152	12/2011	Hunter et al.	N/A	N/A
2012/0179122	12/2011	Eilat et al.	N/A	N/A
2012/0197219	12/2011	Scanlon et al.	N/A	N/A
2012/0304929	12/2011	Ivri	N/A	N/A
2013/0002095	12/2012	Van der Linden	N/A	N/A
2013/0017283	12/2012	Zemel et al.	N/A	N/A
2013/0025038	12/2012	Frey et al.	N/A	N/A
2013/0053042	12/2012	Tanikawa et al.	N/A	N/A
2013/0079733	12/2012	Burt et al.	N/A	N/A
2013/0118619	12/2012	Loth et al.	N/A	N/A
2013/0140225	12/2012	Decock et al.	N/A	N/A
2013/0150812	12/2012	Hunter et al.	N/A	N/A
2013/0152796	12/2012	Pawl	N/A	N/A
2013/0153677	12/2012	Leen et al.	N/A	N/A
2013/0164436	12/2012	Yagi et al.	N/A	N/A
2013/0172830	12/2012	Hunter et al.	N/A	N/A
2013/0206857	12/2012	Ivri	N/A	N/A
2013/0299607	12/2012	Wilkerson et al.	N/A	N/A
2013/0345672	12/2012	Ferreri et al.	N/A	N/A
2014/0088524	12/2013	Marx	N/A	N/A
2014/0113946	12/2013	Abad	N/A	N/A

2014/0157956	12/2013	Date et al.	N/A	N/A
2014/0171490	12/2013	Gross et al.	N/A	N/A
2014/0187969	12/2013	Hunter et al.	N/A	N/A
2014/0214024	12/2013	Eichler	N/A	N/A
2014/0224267	12/2013	Levitz et al.	N/A	N/A
2014/0242022	12/2013	Vehige et al.	N/A	N/A
2014/0249491	12/2013	Ballou, Jr. et al.	N/A	N/A
2014/0257172	12/2013	Yalamanchili	N/A	N/A
2014/0274910	12/2013	Cumberlidge et al.	N/A	N/A
2014/0276054	12/2013	Hossack et al.	N/A	N/A
2014/0285121	12/2013	Balogh et al.	N/A	N/A
2014/0323931	12/2013	Avni	601/46	A61M 11/00
2014/0336596	12/2013	Wochele	N/A	N/A
2014/0336618	12/2013	Wilkerson et al.	N/A	N/A
2015/0018781	12/2014	Rinderknect	604/298	A61F 9/0026
2015/0035180	12/2014	Shen et al.	N/A	N/A
2015/0036219	12/2014	Shen et al.	N/A	N/A
2015/0040891	12/2014	Avni	N/A	N/A
2015/0086397	12/2014	Ma	N/A	N/A
2015/0097050	12/2014	Ciervo	N/A	N/A
2015/0139973	12/2014	Steinfeld et al.	N/A	N/A
2015/0144128	12/2014	Hijlkema et al.	N/A	N/A
2015/0209174	12/2014	Abreu	N/A	N/A
2015/0209178	12/2014	Blakey et al.	N/A	N/A
2015/0238689	12/2014	Shimizu	N/A	N/A
2015/0256730	12/2014	Shen et al.	N/A	N/A
2015/0276994	12/2014	Shen et al.	N/A	N/A
2015/0308421	12/2014	Vogt	N/A	N/A
2015/0328151	12/2014	Ballou, Jr. et al.	N/A	N/A
2015/0352297	12/2014	Stedman et al.	N/A	N/A
2016/0107180	12/2015	Decock et al.	N/A	N/A
2016/0120833	12/2015	Wan et al.	N/A	N/A
2016/0129467	12/2015	Ciardella et al.	N/A	N/A
2016/0199225	12/2015	Ivri	N/A	N/A
2016/0199230	12/2015	Doshi et al.	N/A	N/A
2016/0213866	12/2015	Tan	N/A	N/A
2016/0263314	12/2015	Pardes et al.	N/A	N/A
2016/0296367	12/2015	Ivri	N/A	N/A
2016/0354559	12/2015	Gavini et al.	N/A	N/A
2017/0028626	12/2016	Delrot et al.	N/A	N/A
2017/0136484	12/2016	Wilkerson et al.	N/A	N/A
2017/0138357	12/2016	Kondo et al.	N/A	N/A
2017/0151088	12/2016	Ballou, Jr. et al.	N/A	N/A
2017/0156927	12/2016	Richter et al.	N/A	N/A
2017/0182510	12/2016	Wilkerson et al.	N/A	N/A
2017/0187969	12/2016	Kitamori et al.	N/A	N/A
2017/0274159	12/2016	Gavini et al.	N/A	N/A
2017/0344714	12/2016	Ballou, Jr. et al.	N/A	N/A

2018/0085251	12/2017	Hunter et al.	N/A	N/A
2018/0108275	12/2017	Newberry et al.	N/A	N/A
2018/0116871	12/2017	Hunter et al.	N/A	N/A
2018/0207030	12/2017	Ivri	N/A	B05B 9/0838
2018/0229247	12/2017	Laidler	N/A	N/A
2018/0236466	12/2017	Laidler	N/A	N/A
2018/0297053	12/2017	Buckland et al.	N/A	N/A
2019/0053945	12/2018	Hunter et al.	N/A	N/A
2019/0074086	12/2018	Ballou, Jr. et al.	N/A	N/A
2019/0099071	12/2018	Ehrmann	N/A	N/A
2019/0201656	12/2018	Lacey et al.	N/A	N/A
2019/0217030	12/2018	Burgess et al.	N/A	N/A
2019/0314195	12/2018	Ivri et al.	N/A	N/A
2019/0314196	12/2018	Ivri et al.	N/A	N/A
2019/0314197	12/2018	Ivri et al.	N/A	N/A
2019/0314198	12/2018	Ivri et al.	N/A	N/A
2020/0022416	12/2019	Alarcon	N/A	N/A
2020/0197218	12/2019	Newell et al.	N/A	N/A
2020/0197220	12/2019	Ivri	N/A	N/A
2020/0246182	12/2019	Ivri	N/A	N/A
2020/0281768	12/2019	Quintana et al.	N/A	N/A
2020/0315842	12/2019	Palanker et al.	N/A	N/A
2020/0330267	12/2019	Li et al.	N/A	N/A
2021/0128350	12/2020	Ivri et al.	N/A	N/A
2021/0137732	12/2020	Quintana et al.	N/A	N/A
2021/0220169	12/2020	Ivri	N/A	N/A
2021/0322209	12/2020	Ivri	N/A	N/A
2021/0322210	12/2020	Ivri	N/A	N/A
2022/0039998	12/2021	Ivri	N/A	N/A
2022/0125631	12/2021	Ianchulev et al.	N/A	N/A
2022/0160542	12/2021	Palanker et al.	N/A	N/A
2022/0192874	12/2021	Ivri	N/A	N/A

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
103118642	12/2012	CN	N/A
104146816	12/2013	CN	N/A
104582647	12/2014	CN	N/A
204813955	12/2014	CN	N/A
105351426	12/2015	CN	N/A
107530509	12/2017	CN	N/A
0622035	12/1993	EP	N/A
0622035	12/1998	EP	N/A
1493410	12/2004	EP	N/A
1493410	12/2004	EP	N/A
H08251948	12/1995	JP	N/A
3055480	12/1998	JP	N/A
2007531577	12/2006	JP	N/A

2010071283	12/2009	JP	N/A
2013535250	12/2012	JP	N/A
2018515153	12/2017	JP	N/A
10-1258025	12/2012	KR	N/A
10-2013-0054352	12/2012	KR	N/A
1994020875	12/1993	WO	N/A
1996000050	12/1995	WO	N/A
2000005482	12/1999	WO	N/A
2001046134	12/2000	WO	N/A
2002072169	12/2001	WO	N/A
2006006963	12/2005	WO	N/A
2010078428	12/2009	WO	N/A
2012009706	12/2011	WO	N/A
2013076682	12/2012	WO	N/A
2013090459	12/2012	WO	N/A
2013090468	12/2012	WO	N/A
2013155201	12/2012	WO	N/A
2013158967	12/2012	WO	N/A
2016115050	12/2015	WO	N/A
2016164830	12/2015	WO	N/A
2018136618	12/2017	WO	N/A
2018227190	12/2017	WO	N/A
2019113483	12/2018	WO	N/A
2020010116	12/2019	WO	N/A

OTHER PUBLICATIONS

Abidi et al., “Lifilegrast: A Novel Drug for Treatment of Dry Eye Disease”, Journal of Pharmacology and Pharmacotherapy, 2016, vol. 7, pp. 194-198. cited by applicant

Ali et al., “Glaucoma and Dry Eye”, Ophthalmology, 2009, vol. 116, p. 1232. cited by applicant

Birkhoff et al., “New Devices for Dispensing Ophthalmic Treatments May be the Key to Managing the Life Cycles of Established Products”, 2010, Drug Delivery Technology, vol. 10, pp. 16-21. cited by applicant

Brenton, “CRUK/10/30: TRICON8—Sample collection of ovarian cancer tissues and blood for translational research from patients participating in the CR-UK/MRC ICON8 trial”, 2015, online abstract. cited by applicant

Choi et al., “Generation of Controllable Monodispersed Sprays Using Impulse Jet and Charging Techniques”, Review of Scientific Instruments, 1990, vol. 61, pp. 1689-1693. cited by applicant

Denion et al., “A 5-Minute Interval between Two Dilating Eye Drops Increases Their Effect”, Jul. 19, 2017, Optometry and Vision Science, vol. 94, pp. 838-844. cited by applicant

Electronic Tutorials, “Linear Solenoid Actuator”, 2016 [online]; [Retrieved on Oct. 17, 2022], Retrieved from the Internet [URL: https://www.electronics-tutorials.ws/io/io_6.html]. cited by applicant

Elert, Glenn, “Spherical mirrors”, The Physics Hypertextbook, 2021 [online]; [Retrieved on Oct. 17, 2022], Retrieved from the Internet [URL: <https://physics.info/mirrors/>]. cited by applicant

Gannon, Megan, “The Best Length for Eyelashes, According to Science”, Feb. 24, 2015 [online]; [Retrieved on Oct. 17, 2022], Retrieved from the Internet [URL: <https://www.livescience.com/49934-optimal-length-for-eyelashes-discovered.html>]. cited by applicant

Ianchulev et al., “Pharmacodynamic profile of mydriatic agents delivered by ocular piezo-ejection microdosing compared with conventional eyedropper”, 2016, Ther. Deliv., vol. 7, pp. 751-760.

cited by applicant
Jow et al., “Design and Optimization of Printed Spiral Coils for Efficient Transcutaneous Inductive Power Transmission”, IEEE Transactions on Biomedical Circuits and Systems, 2007, vol. 1, pp. 193-202. cited by applicant
Kent, Christopher, “Getting Meds onto the Eye, 21st Century Style”, Mar. 15, 2013 [online]; [Retrieved on Aug. 27, 2019], Retrieved from the Internet [URL: <https://www.reviewofophthalmology.com/article/getting-meds-onto-the-eye-21st-century-style>]. cited by applicant
Kompella et al., “ISOPT Clinical Hot Topic Panel Discussion on Ocular Drug Delivery”, 2019, J. Ocul. Pharmacol. Ther., vol. 35, pp. 457-465. cited by applicant
Lallemand et al., “Cyclosporine A Delivery to the Eye: A Comprehensive Review of Academic and Industrial Efforts”, European Journal of Pharmaceutics and Biopharmaceutics, 2017, vol. 117, pp. 14-28. cited by applicant
Lindblad et al., “Production of Uniform-Sized Liquid Droplets”, Journal of Scientific Instruments, 1965, vol. 42, pp. 635-638. cited by applicant
Lux et al., “A Comparative Bioavailability Study of Three Conventional Eye Drops Versus a Single Lyophilisate”, Br. J. Ophthalmol., 2003, vol. 87, pp. 436-440. cited by applicant
Macmillan Online Dictionary, “Stream,” n.d. [online]; [Retrieved on Dec. 13, 2018]; Retrieved from the Internet [URL: https://macmillandictionary.com/dictionary/american/stream_1#stream_9]. cited by applicant
Marx et al., “Ophthalmic Squeeze Dispenser: Eliminating the Need for Additives in Multidose Preservative-Free Eyecare Formulations”, 2017, Drug Development & Delivery, vol. 17, pp. 40-44. cited by applicant
Merriam-Webster, “Clamp,” 2019 [online]; [Retrieved on Oct. 25, 2022], Retrieved from the Internet [URL: <https://www.merriam-webster.com/dictionary/clamp>]. cited by applicant
Merriam-Webster, “Collimate,” 2020 [online]; [Retrieved on Oct. 17, 2022], Retrieved from the Internet [URL: <https://www.merriam-webster.com/dictionary/collimated>]. cited by applicant
Merriam-Webster, “Stream,” n.d. [online]; [Retrieved on Dec. 13, 2018], Retrieved from the Internet [URL: <https://www.merriam-webster.com/dictionary/stream>]. cited by applicant
Murube et al., “Classification of Artificial Tears, I: Composition and Properties”, Advanced Experimental Medical Biology, 1998, vol. 438, pp. 693-704. cited by applicant
Murube et al., “Classification of Artificial Tears, II: Additives and Commercial Formulas”, Advanced Experimental Medical Biology, 1998, vol. 438, pp. 705-715. cited by applicant
Oxford Online Dictionary, “Stream,” n.d. [online]; [Retrieved on Dec. 13, 2018]; Retrieved from the Internet [URL: <https://en.oxforddictionaries.com/definition/us/stream>]. cited by applicant
Pronin et al., “Teaching an Old Drug New Tricks: Agonism, Antagonism, and Biased Signaling of Pilocarpine through M3 Muscarinic Acetylcholine Receptor”, 2017, Mol. Pharmacol., vol. 92, pp. 601-612. cited by applicant
Vocabulary.com, “Stream,” n.d. [online]; [Retrieved on Dec. 13, 2018]; Retrieved from the Internet [URL: <https://www.dictionary.com/stream>]. cited by applicant
International Search Report and Written Opinion dated Jul. 16, 2021 in corresponding International Patent Application No. PCT/US2021/027773 (9 pages). cited by applicant

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation patent application of U.S. application Ser. No. 17/233,105, filed Apr. 16, 2021, which claims the benefit of and priority to U.S. Provisional Patent Application No. 63/011,808, filed on Apr. 17, 2020, each of which is hereby incorporated by reference in their entireties.

FIELD OF THE INVENTION

(1) The present invention generally pertains to devices for dispensing fluid medicines and more particularly pertains to such devices that store and deliver ophthalmic preservative-free medicines, specifically configured to increase the ease of use and enhance patient compliance with dosing instructions for the medicine.

BACKGROUND

(2) Ease of dispensing fluid medicines and compliance with dosing instructions are primary concerns with all patients. In particular, preservative-free dispensing bottles such as ophthalmic squeeze dispensers typically require greater actuation force due to a valve mechanism that seals the dispensing nozzle to prevent bacterial ingress and contamination. Such system requires much higher pressure to operate and hence much higher squeeze force is required. In addition, prior art dispensing bottles dispense only in an upside-down orientation which require an inconvenient head maneuver and which together with higher actuation force further increases the inconvenience.

(3) Dispensers of the kind in question are known from the prior art, for example from U.S. Pat. Nos. 6,095,376, 9,676,525, US 2014/0336596, US 2016/0107180, U.S. Pat. Nos. 9,238,532, 8,056,766, 8,863,998, and 10,105,720. The dispenser shown in US 2014/0336596 comprises an outlet channel which connects the liquid reservoir to the outlet opening through an outlet valve which is arranged in the outlet channel and which opens when the bottle is squeezed and pressure is generated. Such preservative-free squeeze bottles typically require about 25-28N of squeeze force (*Ophthalmic Squeeze Dispenser—Drug Development and Delivery* October 2017 Vol. 17 No. 7 page 40). Elderly patients, or other patients lacking enough strength and/or dexterity in their hands, often experience problems dispensing medicine from such bottles.

(4) This work provides preservative free ocular dispensing device that can be held horizontally, or in any convenient orientation while the actuation is done effortlessly by an electrical switch. This provides a cost effective solution that is consistent with standard drug packaging processes.

SUMMARY

(5) Multi-dose preservative-free ocular fluid delivery devices are provided. The fluid delivery device includes a fluid dispensing system and a fluid package for storing a liquid therein and supplying said liquid to the dispensing system. The dispensing system comprises an elongated chamber which includes a check valve which defines a frontal closure to the chamber. The valve is normally closed and hermetically seals the chamber. In this work the chamber includes a vibration motor that induces oscillations to the chamber and to the fluid within. The oscillations of the chamber impart momentum to the fluid stored in the chamber which in turn imparts force that cyclically opens the valve to dispense streams or liquid droplets. Fluid is dispensed only when the motor oscillates while otherwise the valve is hermetically closed.

(6) The check valve can include a flexible plate which includes a conical aperture that extends through its thickness, the valve can further include a stationary spherical member that engages tangentially with the inner wall of the conical aperture to create a hermetic sealed closure. The plate can be made of elastomer that has a modulus of elasticity ranging between 0.1-1.2 GPa. The circumference of the plate can be attached to the chamber by a retaining ring that engages with the chamber in an interference fit to create the hermetically sealed closure.

(7) The conical aperture extends through the thickness of the plate such that droplets are dispensed

through the smaller opening of the aperture while the larger side of the aperture is in fluid communication with the chamber.

(8) The spherical member may include an antibacterial coating which covers the area of the spherical member that is between the tangential engagement line and the small opening of the aperture.

(9) The vibrational motor oscillates the chamber and the fluid within the chamber. Consequently, cycles of hydrodynamic pulses are generated causing the valve to cyclically open and dispense fluid. Here this phenomenon is characterized by oscillatory interactions between the valve and the surrounding fluid. The hydrodynamic force generated by the momentum of the fluid opens the valve and allows fluid flow through the aperture.

(10) Fluid is dispensed only when the hydrodynamic force is sufficiently high to deform the aperture while otherwise the aperture hermetically seals the chamber. The system prevents ingress of microorganism into the chamber allowing storage of preservative free pharmaceutical. This work provides an electrically operated preservative-free dispensing system that is convenient and cost effective.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. 1A is a cross sectional view of an embodiment of the invention.

(2) FIG. 1B is a detail cross sectional view of the embodiment of FIG. 1A.

(3) FIG. 1C is a 3D view of the embodiment of FIGS. 1A-B.

(4) FIG. 2A is a side view and partial cross section view of an embodiment of the invention in operation.

(5) FIG. 2B is a detail cross section view of the example of FIG. 2A.

(6) FIG. 3 illustrates a perspective view of a vibration motor as used in embodiments of the invention.

(7) FIG. 4A is a frontal view of an embodiment of the invention.

(8) FIG. 4B is a side view of the embodiment of FIG. 4A.

(9) FIG. 4C is a detail cross sectional view of the embodiment of FIG. 4A.

(10) FIG. 5A is a frontal view of an exemplary device enclosed in a housing.

(11) FIG. 5B is a side view of the example of FIG. 5A.

(12) FIGS. 6A-B are perspective views of a device having a housing that includes a swivel cover.

DETAILED DESCRIPTION

(13) This work describes dispensing devices and methods for delivery of preservative-free solutions or suspensions for ocular administration of ophthalmic drugs. The dispensing devices include a droplet ejecting system that is fluidly connected to an ampoule package containing a liquid to be dispensed. The droplet ejecting system includes a chamber having a check valve that defines a front closure to the chamber. The dispensing system further includes a vibration motor that oscillates the chamber and induces hydrodynamic pulses which consequently causes the valve to cyclically open and eject fluid droplets. The valve is normally closed and hermetically sealing the chamber. The valve opens exclusively in response to hydrodynamic pulses induced by the oscillation of the chamber. In this way fluid is dispensed only when the device is actuated while otherwise the aperture hermetically seals the device and prevents ingress of bacterial and microorganisms thereby allowing storage of preservative-free pharmaceutical formulation. The use of a vibration motor further enables convenient and cost effective, electronically controlled administration.

(14) FIG. 1A and FIG. 1B illustrate a side view and an enlarged partial section view, respectively, of a fluid delivery device **100**. Delivery device **100** includes a fluid reservoir **102** and dispensing

system **104** connected to each other in fluid transmission relationship through passage **106**. Dispensing system **104** includes a fluid chamber **108** and also includes a check valve including aperture plate **110** which provides frontal closure to the chamber **108**. Referring to FIG. **1B** it can be seen that aperture plate **110** includes a conical or tapered aperture **116** (shown inside a dotted circle for clarity) that extends through its center thickness and has a large inlet opening **116a** in fluid communication with chamber **108** and a smaller exit opening **116b** through which fluid droplets will be dispensed. Aperture plate **110** can be made of a flexible elastomer such as silicone rubber, VersaFlex, manufactured by VersaFlex Incorporated Kansas City, Kansas USA. Other elastomers with Young modulus of elasticity value between 0.5 GPa to 2 GPa can also be used. Dispensing system **104** further includes a stationary spherical member **114** that tangentially engages in pressure transmission relationship with the inlet opening **116a** of the conical aperture providing hermetically sealed closure. In a preferred embodiment spherical member **114** is made of high density polyethylene (HDPE) that is harder than silicone thereby creating a tight closure as it engages with the softer aperture plate **110**. Spherical member **114** preferably engages in pressure transmission relationship with the conical aperture **116a** with a preload force of between 0.01N and 0.05N. The combination of aperture plate **110** and spherical member **114** provides a check valve as described above. Spherical member **114** is supported by pin member **112**. Member **114** can have a shape other than spherical, since any shape capable of forming a good seal with aperture plate **110** can be used.

(15) Aperture plate **110** can be retained to dispensing system **104** by a retaining ring **130**, thereby creating a hermetically sealed closure.

(16) Dispensing system **100** includes a venting tube **126** that is configured to equalize the pressure inside container **102** as the fluid is dispensed from the device. The opening **134** of venting tube **126** is extended above the fluid level **132** at any orientation that the device is held. Vent tube **126** can be connected via a 0.22 micron filter **128** to assure that the air that enters the device is sterile.

(17) Dispensing system **100** further includes a vibration motor configured to oscillate chamber **108** and the fluid within the chamber. Here this motor is schematically shown as eccentric mechanical load **118** which vibrates the assembly as described when it is rotated by the motor (motor not shown in FIGS. **1A-B**, but described below in connection with FIG. **3**).

(18) FIG. **1C** is a 3D view of the embodiment of FIGS. **1A-B**.

(19) FIGS. **2A-B** illustrate the response of the dispensing system **104** to oscillations. Vibrations of the motor induce oscillation to the body of dispensing system **104** which in turn generates fluid momentum within the chamber due to the dynamical interaction of the fluid with the solid structure of chamber **108**. This phenomenon is often referred to as Solid-Fluid-Interaction (SFI). The momentum of the moving fluid exerts force that flexes the aperture plate **110** outwardly in the direction indicated by arrows **206a** and **206b** causing the aperture plate **110** to disengage from spherical member **114** thereby opening a fluid flow passage as indicated by the arrows **208a** and **208b** and ejection of fluid droplets **210**.

(20) Dispensing device **100** is supported by a flexible beam **122** or other structural embodiments which allows it to oscillate freely, as schematically shown by motion excursions **202** and **204**. Preferably the spring constant of the beam **122** is 0.05 N/mm to 0.5 N/mm. For example, beam **122** can be formed by fabricating a slot **124** in support structure **120** such that the resulting beam **122** has a thickness suitable for providing a spring constant as recited above.

(21) FIG. **3** illustrates an exemplary vibrational motor **302**. The DC Motor **302** of this example has a cylindrical body with a diameter of 4 mm and total length of 17 mm. An eccentric flywheel **118** is attached to the motor shaft. The flywheel has a mass of 1.7 grams with a center of mass about 0.7 mm from the center of rotation. The motor receives 4.5-12 VDC and rotates at 6000-12000 RPM generating centrifugal force of 0.3N at a rotation speed of 8000 RPM. Other DC motors that generate centrifugal force of 0.1-1N and rotation speed of 1000-50000 RPM can be used. The motor can be controlled by a timer circuit which is set to provide an ON time as required to deliver

a dose of 8-12 micro-liter. The actuation ON time is 60-200 ms depending on the rheology of the fluid in use. A timer circuit which incorporates a 555 timer IC or a microprocessor-based timer with a 12 volt battery such as A23 alkaline battery may be used.

(22) FIGS. 4A-C illustrate an alternative preferred embodiment of dispensing system **400**. FIG. 4A illustrates a frontal view and FIG. 4B illustrates a side view of dispensing system **400**. Dispensing system **400** includes a concave mirror **402** which assists in aligning device **400** and fluid stream **210** to the eye of the user. In use, dispensing device **400** is positioned in front of the eye such that the image of the eye appears sharply and in the center of mirror to the user. At this point the device is properly positioned in term of the distance of the eye from the nozzle **404** and its angular orientation relative to the eye. Upon actuation, stream **210** will be deposited precisely on the corneal surface of the eye. Device **400** preferably includes a 0.22 micron air filter configured to filter the vented air that flows into device **400**, as in the previous example.

(23) Device **400** includes a check valve having an aperture plate **406** with a conical aperture **116** that extends through its thickness. The check valve further includes a spherical member **114** that tangentially engages with the inlet opening of the conical aperture **116**. In this example, the check valve also includes a compression spring **408** configured to force aperture plate **406** against spherical member **114**. In this way a tight seal is created along the engagement line **114a**, thus creating a tight and hermetic closure.

(24) Spherical member **114** can be partially covered with an antimicrobial coating, specifically in the area of spherical member **114** that is not in contact with fluid in the chamber. The coated area thus extends between the engagement line **114a** and the outlet of the conical aperture (i.e., to the left of **114a** on FIG. 4C)). Examples of antimicrobial coatings include silver in metallic form, silver alloy or a non-metallic material that contains silver including salts of silver chloride and silver sulfadiazine. Other optional material includes biguanide derivatives, chlorhexidine diacetate and chlorhexidine digluconate.

(25) FIGS. 5A-B illustrate frontal and side views, respectively, of a dispensing device **400** packaged in a housing **502**. Housing **502** provides a convenient enclosure to the dispensing system **400** that was described in relation in FIGS. 4A-C and is shown with dashed lines here. Housing **502** includes an electrical circuit (not shown) and a 12 volt battery (e.g., type A23) **506**. The circuit controls the dispensing period such that a dose of 8-12 microliter is delivered to the ocular surface of the eye. Momentary switch **504** can be used to activate the device **400** such that a stream of droplets is ejected from the aperture as described earlier. FIGS. 5A-B also illustrate the mirror **402** which is visible on the front side of the device. Such a housing can also be used for a dispensing device **100** as described above.

(26) FIGS. 6A-B illustrate some preferred features of housings that can be used with embodiments such as device **100** and device **400** as described above. In this example, the housing includes a swivel cover **602** that covers dispensing nozzle **606** during periods of non-use. Swivel cover **602** provides a means to prevent bacterial contamination on the external areas where a residual fluid may be left following each use. Such residual fluid may contaminate subsequent stream as it is dispensed through the nozzle. Swivel cover **602** preferably includes a flexible member **604** that includes a surface **604a** that is covered with antimicrobial coating. Surface **604a** engages with the outlet opening of nozzle **606** when the swivel cover **602** is closed as illustrated in FIG. 6B. In this way antipathogen action is achieved. Alternatively, organic dyes with antiseptic action may also be used. For example, toluidine blue, methylene blue, gentian violet and acridine and related active substances such as acridine orange and acridine yellow as well as ethacridine lactate. Germicidal polymers such as polyhexanide are also possible. Materials that contain additives which contain metal-organic substances with an ionizing effect can also be used. Such additives are available from SteriOne GmbH Berlin, Germany. Examples of antimicrobial coatings that can be used on surface **604a** include silver in metallic form, silver alloy or a non-metallic material that contains

silver including salt of silver chloride and silver sulfadiazine. Other optional material includes biguanide derivatives, chlorohexidine diacetate and chlorohexidine digluconate.

Claims

1. A method for delivering fluid to an eye of a patient, the method comprising: using an actuator to oscillate a chamber so as to generate fluid momentum from the fluid therein; allowing the fluid momentum to disengage an aperture from a pin member disposed within the chamber to eject the fluid through the aperture; and reengaging the aperture with the pin member to seal the aperture after the fluid is delivered.
 2. The method of claim 1, wherein the aperture and the pin member are hermetically sealed when the fluid is not being ejected.
 3. The method of claim 1, wherein the aperture is a flexible aperture.
 4. The method of claim 1, further comprising: biasing the aperture against the pin member via a compression spring.
 5. The method of claim 1, wherein the reengaging step further comprises preventing an ingress of bacteria or contaminants as a result of the engagement between the aperture and the pin member.
 6. The method of claim 1, further comprising: providing a mirror such that when the eye of the patient is in-focus and centered in the mirror, ejection of the fluid through the aperture will be delivered accurately to the eye.
 7. The method of claim 1, further comprising: equalizing a pressure of the fluid as it is ejected through the aperture.
 8. The method of claim 7, wherein the equalizing step includes a vent in fluid communication with the chamber and ambient air and introduces air into the chamber as the fluid is ejected.
 9. The method of claim 8 further comprising: filtering the air to ensure the air is sterile before introduction into the chamber.
 10. An apparatus for delivering a fluid to an eye of a patient, the apparatus comprising: a) a reservoir to accommodate the fluid; b) a liquid dispenser in fluid communication with the reservoir, the liquid dispenser having a check valve through which the fluid can be selectively dispensed; c) an actuator positioned to vibrate the liquid dispenser and generate fluid momentum from the fluid therein to open the check valve and eject the fluid therethrough; and wherein the check valve is a pin member engaging an aperture.
 11. The apparatus of claim 10, wherein the actuator includes a vibration motor having an eccentric mechanical load relative to an axis of rotation of the vibration motor.
 12. The apparatus of claim 10, wherein the check valve includes the aperture which is tapered to be narrower at an outlet of the aperture than at an inlet of the aperture.
 13. The apparatus of claim 10, further comprising a vent to equalize a pressure of the fluid as the fluid is ejected.
 14. The apparatus of claim 13, wherein the vent includes a filter configured to ensure that, as ambient air is introduced, the ambient air that enters is sterile.
 15. The apparatus of claim 10, further comprising a concave mirror configured to provide an in-focus image of the eye of the patient to the patient when the liquid dispenser is positioned properly to deliver the fluid to the eye of the patient.
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